# PALEOMAGNETISM OF THE OLDEST RED BEDS AND THE DIRECTION OF THE LATE APHEBIAN POLAR WANDER RELATIVE TO LAURENTIA

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Abstract. The direction of the apparent polar motion relative to Laurentia for the interval 2.2 to 1.9 b.y. ago is the subject of a controversy owing to difficulties in determining the age relationships and age of the complex magnetization of those very old rocks. Three cleaning treatments (thermal, chemical and alternating field) were used to study the fossil magnetization of 104 specimens (8 sites), of 2.3 b.y. old red argillites of the Huronian Supergroup. The results indicate that the magnetization was acquired during the initial process of magnetization. The mean direction 337°,+52° (N=8; $\alpha_{95}$=8°) yields a pole at 158°E, 67°N ($\delta p$ = 8°; $\delta m$ = 11°). This and other recently published results indicate that the apparent polar motion relative to Laurentia for the 2.3 to 1.9 b.y. ago was mainly latitudinal and from north to south.

## Introduction

Until recently, owing to the scarcity of paleomagnetic and age determination results, it has been difficult to construct the polar path beyond 1.9 b.y ago. As more paleomagnetic and radiometric results become available, it appears that the apparent polar motion relative to Laurentia for the interval 2.2 to 1.9 b.y. ago is latitudinal. However, both a south to north (Spall, 1971; Symons, 1971, 1975; Irving et al, 1972; Fahrig and Chown, 1973) and a north to south (Irving and Lapointe, 1975; Pullaiah and Irving, 1975; McGlynn et al, 1975) motions have been proposed (an east to west motion has also been suggested by Larson et al, 1973). These competitive ideas arise from the problem in establishing the age relationships of the igneous bodies emplaced during that time interval and the age sequence of the different phases of magnetization to which those old rocks have been subjected. Thus the early phases of magnetization acquired during formation of the rock are not easily identified and sometimes may not even be uncovered without complex demagnetization techniques (Park, 1975; Irving and McGlynn, 1975).

The world's oldest red beds are found in the Huronian Supergroup (Roscoe, 1973), Ontario, Canada. A Rb-Sr isochron age of 2,288±87 m.y. has been obtained (Fairbairn et al, 1969) from graywacke and argillite of the Gowganda Formation of the Cobalt Group. These beds are therefore about 100 m.y. older than the upper limit of the time interval in question and are intruded by some of those igneous bodies. If the initial magnetization of these beds can be recovered, its pole should provide us with the sense of the apparent latitudinal motion and resolve a problem which is vital to magnetostratigraphy and the study of the structural history of the lower Proterozoic.

## Results

There are good exposures of gently dipping (≃7°) red argillites of the Firstbrook Member of the Gowganda Formation in the Cobalt area (280.4°E, 47.4°N) and 56 core samples (2.5 cm dia) were collected from 8 sites for paleomagnetic studies. The sites were sampled over a distance of 20 km and a stratigraphic thickness of more than 30m. The cores were sliced into 104 specimens (2.2 cm height) and subjected to thermal (56 sp), chemical (25 sp) and alternating field (a.f.) (23 sp) cleaning treatments. In typical specimens (Fig. 1), more than 70% of the magnetization has blocking temperatures above 650°C and nearly half of the magnetization remains after demagnetization in zero field at 675°C; 65 to 80% of the magnetization remains after cleaning in alternating fields of 1500 gauss (peak) and the median destructive field is about 2000 gauss. With such high blocking temperatures and high remanent coercive forces, it is most probable that hematite is the carrier of magnetization. The chemical treatment consisted of immersing the specimen in hydrochloric acid for a given time (Fig. 1), washing in water, storing in zero field for 2 days, measuring, and re-immersing in fresh acid. The acid strength was 8 N for the first 370 hours and 10 N thereafter. These argillites are fine-grained and massive, so that the rate of penetration of acid is very slow. Specimens were sliced at different stages of leaching for observation and there is a close correlation between the percentage of remaining intensity and the percentage (≃40% after 2300 hr) of unleached material. It then appears that the chemical treatment has leached out most (if not all) of the magnetic carriers from the leached portion while leaving the magnetization of the unleached portion practically untouched.

The direction changes occurring during cleaning are small but different for each of the three cleaning treatments. Thermal demagnetization causes a systematic decrease in inclination of 8 to 10°, a.f. demagnetization up to 1500-2000 gauss produces a decrease of about half that amount, and chemical leaching does not produce any appreciable changes. Polished sections of unleached and leached specimens were examined. The packing density of these beds is very high; after 2300 hr acid penetration is only 2 to 3 mm. No magnetic carrier can be detected in the leached section. In the unleached specimens, both the red pigmentation and a small percentage

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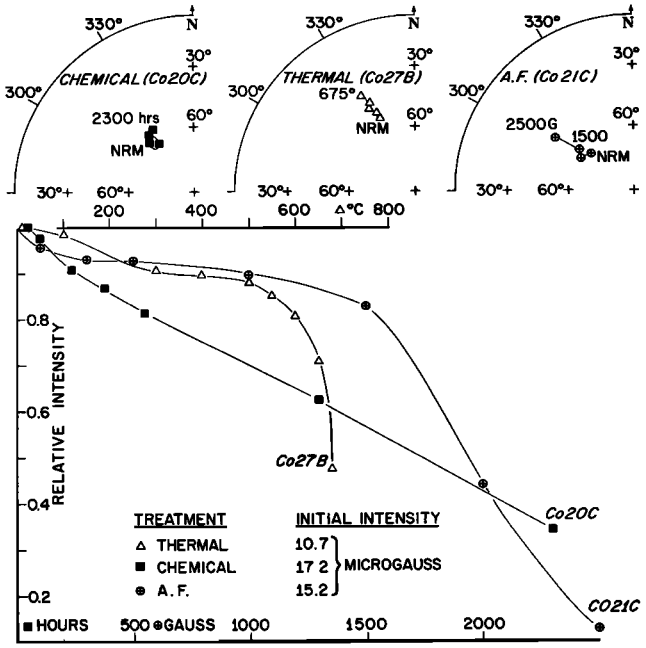



Figure 1: Results obtained from thermal, chemical and alternating field (a.f.) cleaning treatments showing a magnetization with high blocking temperatures and very high remanent coercive forces. The intensity on the chemical decay curve after a certain period of leaching is roughly proportional to the ratio of unleached material. The directions (all positive) have been corrected for bedding tilt.

(≃.05%) of hematite particles are homogeneously distributed throughout the rock. The specimens examined are fracture-free and there is no evidence of any alteration which could have taken place after final lithification. These observations are in accordance with the fact that none of the cleaning treatments shows any indication of a secondary magnetization. It is therefore concluded that the magnetization of these red beds occurred during deposition and/or diagenesis and was completed by lithification time.

The results are consistent with the process of magnetization proposed previously for Carboniferous red beds (Roy and Park, 1974); that is, the magnetization was acquired during a continuing and lengthy process of magnetization which commenced with deposition of magnetic particles followed successively by hematite accretion, filling, and cementing during diagenesis. In that study, comparison of thermal and chemical work on specimens (which were more porous than the argillite studied here) of the same core sample showed that the magnetization carried by the red pigmentation and acquired during the late phases of the magnetization process had blocking temperatures distributed over a wide range (20° to about 650°C) while the magnetization acquired during the early phases had blocking temperatures near the Curie point (675°C) of hematite. The similarity of the thermal results obtained in both (the Carboniferous and this) studies suggests that the fossil magnetization of the argillites was acquired over the entire process of magnetization and that the magnetization remaining after demagnetization at 650°C was acquired during the early phases of the magnetization process. The magnetization is little affected by a.f. demagnetization below 1500 gauss indicating high remanent coercive forces for all phases. However, the decrease in inclination produced by thermal demagnetization also occurs (on a smaller scale) in fields of 1500 - 2500 gauss (Fig. 1.) indicating that the early phases of magnetization have very high remanent coercive forces. Although the magnetic carriers of the different phases of magnetization may depend on the grain size and/or solubility in acid, those phases cannot be separated by leaching in HCl. This is attributed to the slowness of penetration rate which allows enough time for the acid to destroy all magnetic carriers (regardless of the grain size and solubility) before proceeding to attack any of the grains buried deeper within the specimen. This indicates that the remaining unleached material (still) represents the mean direction of all phases of the fossil magnetization and explains the fact that no appreciable direction changes occur during chemical leaching.

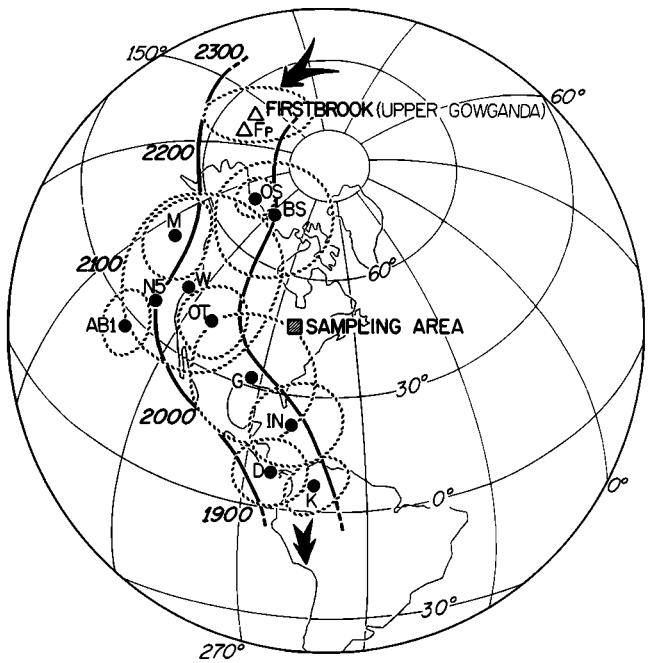



Figure 2: A 20° proposed apparent polar path for Laurentia for the interval 2,300-1,900 m.y. ago. The 20° width is to emphasize the uncertainties associated with the determination of paleopoles of that age. Except for the FIRSTBROOK and the N poles, the other poles listed in Table 1 have been redrawn from McGlynn et al (1975) and the same notations are used. Error circles or ellipses at P=0.05 have been drawn. The Firstbrook triangle is regarded as the pole position during the early phases of magnetization. The $F_p$ pole is probably a fair estimate of the mean pole position during all phases of the magnetization process and is consequently younger than the Firstbrook pole. This is in agreement with the proposed apparent polar motion from north to south.

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Table 1. Paleomagnetic and Radiometric Results for Poles of Figure 2.

| Index | Rock Unit | Age in m.y. ( )† | A,B ‡ |
|---|---|---|---|
| F, $F_p$ | Firstbrook Member | 2287±88 (R) | 1, 14 |
| N* | Gowganda Formation | 2287 88 (R) | 2, 14 |
| OS | Otto stock | 2160 80 (R) | 3, 15 |
| BS | Big Spruce complex | 2170 40 (R) | 4, 16 |
| M | Mugford basalt | 2100 (R) | 5, 17 |
| W | Wind River dikes | 2060 (K) | 6, 06 |
| N5 | Nipissing diabase | 2150 50 (R) | 7, 14 |
| OT | Otish gabbro | 2000 (S) | 8, 08 |
| AB1 | Abitibi dikes | 2147 68 (R) | 9, 18 |
| G | Gunflint Formation | 2000 (S) | 10, 10 |
| IN | Indin dikes | 2093 86 (R) | 11, 18 |
| D | Dabawnt Group | 1830 26 (R) | 12, 12 |
| K | Kahochella Formation | 1873 13 (R) | 13, 13 |

†Radiometric method for age cited: (R) RB/Sr; (K) K/Ar; (S) stratigraphic assignment.
‡References A and B refer to the source of the paleomagnetic and radiometric information respectively:1-this paper;2-Symons,1975;3-Pullaiah and Irving,1975;4-Irving and McGlynn,1975; 5-Murthy and Deutsch, 1972;6-Spall,1971;7-Symons,1971;8-Fahrig and Chown,1973;9-Larochelle, 1966;10-Symons,1966;11-McGlynn and Irving,1975;12-Park et al, 1973;13-McMurray et al,1973; 14-Fairbairn et al, 1969;15-Bell and Blenkinsop,1975;16-Martineau and Lambert,1974;17-Barton,1973; 18-Gates and Hurley,1973.
*The paleomagnetic result (N) was obtained from red beds in the Blind River-Elliot Lake area after a.f. demagnetization at 200 gauss (one specimen per site was demagnetized up to 800 gauss). Without thermal, chemical and/or high a.f. work, one cannot assume that the magnetization is initial. In fact, a F-distribution test for comparison of precision (K) (Watson, 1956) shows the magnetization to be post-folding (the site-means K without and with correction for folding are 42 and 15 respectively yielding a K ratio of 2.8 which is significant at the 99% level). The nearness of poles OS (2,160 m.y.) and BS (2,170 m.y.) indicates the possibility that N represents a remagnetization at that time; that is, about 100 m.y. after deposition.

The Firstbrook direction obtained from the 8 site means after cleaning at 650°C is 337°,+52° with a circle of confidence $\alpha_{95}$=8° (site given unit weight). The circular standard deviation is 12° and comparable to the 13° calculated as that of the present field at that latitude (Creer, 1962). That direction is then regarded as a good estimate of the mean direction of the Aphebian magnetic field during the early phases of the magnetization process. The mean direction ($F_n$) obtained by averaging the 2300 hr chemical and the 100°C thermal results (hence removing possible viscous components) is 333°,+55° and is probably a fair estimate of the mean direction of the magnetization acquired during the whole of the magnetization process. The time span of the magnetization process cannot be determined directly. However, results from Late Cenozoic red beds (Larson and Walker, 1975) indicate that the process can take more than 2 to 5 m.y. for completion; results from Carboniferous red beds (Roy and Park, 1974) indicate that it is completed in less than 35 m.y.. On the basis of the present evidence, it is reasonable to assume that completion of the process is of the order of a few tens of m.y.. This means that, if the dating of Fairbairn et al (1969) is accepted, the magnetization of the Firstbrook is probably about 2250 m.y. old.

A path of apparent polar wander relative to Laurentia for the interval 2,300 to 1,900 m.y. ago is shown in Fig. 2. The paleomagnetic poles and relevant information with references are listed in Table 1. The magnetization of the Firstbrook is older than that of any other rock unit listed in Table 1 and this result extends previous polar paths. The FIRSTBROOK pole (and also the relatively younger $F_p$) indicates that the Late Aphebian polar wander relative to Laurentia was mainly latitudinal and the direction from north to south.

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